

Practical Applications of Cyber security: A Research Paper on Knowledge Graphs for Real- World Security Solutions

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Abstract— The realm of cybersecurity is riddled with intricate and ever-evolving cyber-attack activities, presenting formidable obstacles to cybersecurity professionals. By incorporating knowledge graphs into this field, one may help with threat identification and situational awareness by visualizing the intricate cybersecurity landscape. But creating these knowledge graphs might add noise and conflicts, which could result in incorrect conclusions. The goal of this work is to address the difficulties involved in creating and assessing cybersecurity knowledge graphs. To offer a more flexible framework appropriate for different data properties, we improved the current cybersecurity ontology and put together a cybersecurity knowledge graph dataset (CS13K). The cybersecurity knowledge graph is made with Neo4j. In addition, we developed an AttTucker model to evaluate knowledge graph quality, which is based on Transformer architecture. In order to reduce the dimensionality of knowledge embeddings and capture latent links among entities and relations, this model uses multiple self-attention heads, producing assessment results that are similar to those of higher dimensionally embedding. Additionally, in the AttTucker model-based evaluation procedure, we incorporated path-level data from knowledge graph entities. Our method significantly improves F1 value and accuracy, surpassing current models in knowledge graph quality evaluation tasks, according to an experimental investigation using cybersecurity and general datasets.

Keywords—Knowledge graph • Cybersecurity • Ontology construction • Attention mechanism • Quality assessment

I. INTRODUCTION

A plethora of priceless cybersecurity knowledge has spread over the internet in recent years, dispersed among numerous platforms such as vulnerability databases, security forums, and information archives. Despite their abundance, these fragments of cybersecurity knowledge remain largely disconnected, failing to leverage the full potential of big data insights. Moreover, the ever-evolving nature of cyber threats demands more than passive defense mechanisms; proactive analysis of cybersecurity data has become imperative [1]. Enter knowledge graphs, hailed as a potent solution amidst the complexity and diversity of cyber threats. These graphs serve as dynamic repositories of interconnected knowledge,

automatically unveiling associations between threat entities, facilitating cybersecurity situational awareness, risk assessment, and even aiding in attack path visualization during forensic analysis. By mining these associations, cybersecurity professionals gain deeper insights into vulnerabilities and attack vectors, empowering them to fortify defense strategies and bolster overall cybersecurity posture. Despite existing general-purpose ontologies, their applicability often falls short due to varying focuses on cybersecurity elements. To bridge this gap, we present a novel cybersecurity ontology tailored to accommodate diverse data sources and support the development of comprehensive and accurate knowledge graphs [2,3].

While knowledge graphs have proven invaluable across numerous domains, ensuring the integrity of their content is paramount. Inaccuracies in these graphs have the potential to spread to more advanced applications, resulting in erroneous inference and judgment. Yet, the construction of knowledge graphs inevitably introduces noise and conflicts, necessitating robust quality assessment mechanisms. Traditional manual inspection methods are impractical for large-scale knowledge repositories, underscoring the need for automated quality evaluation techniques. To improve cybersecurity knowledge graphs' efficacy and dependability in practical applications, we offer a novel method for assessing their quality in response [1-2].

The subsequent sections of the paper are structured as follows:

"Related Works" discusses relevant literature.

"Methods" provides an overview of the proposed model and describes how the CS13K dataset was created. It also describes the methodology used to generate the cybersecurity knowledge graph. "Results and Discussion" presents experimental findings, with a comprehensive evaluation and analysis of the model.

A summary and recommendations for future study directions are included in the paper's conclusion.

II. LITERATURE REVIEW

Our aim was to offer a comprehensive overview of the existing literature and advancements in these key areas of study within the cybersecurity field. Research on cybersecurity knowledge graph datasets has been scarce because cybersecurity data is very sensitive and specialized, and because there are few large-scale specialized corpora. Table 1 shows how Robert et al. [4] used automatic annotation techniques to create a cybersecurity entity recognition dataset. They used the "BIO" labeling scheme and 15,192 cybersecurity utterances [5-7]. This 498,000-word dataset yielded an F1 score of 75.05% using Bi-LSTM-CRF, which was the basis for entity recognition study.

Rastogi et al. [8] collected data from 1100 threat intelligence reports using automated extraction methods to generate the MT40K malware knowledge graph dataset. There are 40,000 triples in this dataset, consisting of 27,354 items and 34 different types of relations. However, due to automated extraction, the accuracy of the dataset's triples is not optimal. The OSINT cybersecurity information extraction dataset was painstakingly put together by Guo et al. [7], who combined information from a range of sources, including hacker forums, security advisories, vulnerability databases, and APT reports. Nevertheless, the dataset is not publically accessible due to the significant internal annotation work. A recent field of research is the application of deep learning and natural language processing techniques to cybersecurity issues. However, the absence of large-scale cybersecurity datasets—which is made worse by the low quality of the few publicly available datasets—remains a significant barrier to the advancement of deep learning-based cybersecurity research.

The foundational step in creating a cybersecurity knowledge graph is creating an ontology. But because they were designed with different priorities in mind, current cybersecurity ontologies frequently provide difficulties for thorough integration across many data sources [9]. The topic of cybersecurity has seen the emergence of numerous ontologies. An ontology for cybersecurity was created by Iannacone et al. [11] and includes 115 properties and 15 entity categories. The Universal Cybersecurity Ontology (UCO), which links several cybersecurity standards and ontologies including CVE, CCE, CVSS, and others, was later developed by Syed et al. [12] building on this work. The STIX format is compatible with the UCO. The eight main entity categories that make up UCO are Attacker, Attack-Pattern, Consequences, and Indicator. Additionally, the Security Asset Vulnerability Ontology (SAVO) was developed by Vorobiev et al. [13] to characterize the characteristics, ideas, and connections between network threats, vulnerabilities, and risks. Pingle et al.'s [14] addition of crucial classes including Campaign, Tool, and Course-of-Action, as well as the enlargement of relations, improved the UCO cybersecurity ontology and its extension. Knowledge graphs are currently a hot topic in research in a variety of fields, including recommendation systems, question-answering systems, and semantic search [15, 16].

III. SYSTEM IMPLEMENTATION

Intelligent search and recommendation systems frequently employ general knowledge graphs, which highlight a wide variety of knowledge. Notable instances of generic knowledge repositories are Wikidata, YAGO,

DBpedia, and Freebase. Conversely, inside particular knowledge domains, domain-specific knowledge graphs highlight specialized knowledge and cater to particular query and analytic needs. Applications for these specialized graphs are widely used in fields including e-commerce, healthcare, finance, and academics. Cybersecurity issues are complex and include talking about attack vectors, security flaws, and related countermeasures in many contexts. These subjects have been thoroughly studied by academics [17], with some using federated learning strategies to thwart cyberattacks [18]. Consolidating a wide range of cybersecurity data sources into knowledge that can be understood semantically is made easier with the help of knowledge graphs integrated into cybersecurity. Cygraph helps with things like identifying attack surfaces and figuring out how to defend against attacks in order to protect important assets [19].

The purpose of knowledge graph quality assessment is to assess a knowledge graph's knowledge content quantitatively. Conversely, detection strategies that rely on the knowledge graph's internal data usually model the graph and identify relationships between items using random walk approaches. A high probability of links between things in the knowledge graph is shown by multiple paths between them. To find false entities and relations, several researchers have used knowledge representation learning techniques. Its ability to capture both local and global structural information for triple confidence assessment was shown by experimental findings. It enables users to build quality assessment functions according to particular business objectives or to calculate the final score by carefully weighing several evaluation techniques. Using the FB15K dataset, Jia et al. [20] successfully evaluated triple confidence in knowledge graphs by evaluating the triple confidence from entity, relation, and global viewpoints. Using automated machine learning (AutoML) technology, Zhang et al. [21] created score systems appropriate for the knowledge graph triple classification problem.

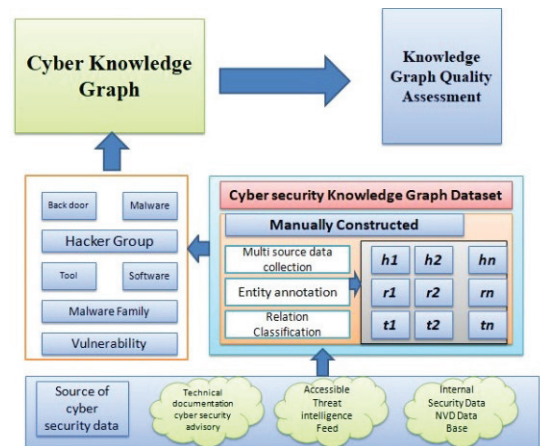


Fig. 1. Structure for building cybersecurity knowledge graphs and evaluating their quality

IV. DATA SET

The technical and sociological facets of cyberattack incidents are presented in cybersecurity threat reports as unstructured text. Extracting important information from these reports is the first step in creating a cybersecurity knowledge graph, and it's not an easy task. The cybersecurity knowledge graph datasets that are now available as open-

source resources were gathered via automatic extraction, and as a result, the accuracy of the triples is low, not reflecting the real state of cybersecurity. Security personnel with strong cybersecurity subject expertise. This dataset is known as CS13K (Cybersecurity 13K). We'll keep updating CS13K to reflect the real state of cybersecurity and broaden its reach. The twelve categories of relations are: indicate, mitigates, targets, use, associate, belong To, cause, exploits, has Attack Location, has Attack Time, has Characteristics, and has Vulnerability. With 13,027 triples overall and matching documents and words, the dataset serves as a data source for upcoming research projects in few fields. Examples of relationships with the notation "belong To" include those between Threat Actor and Hacker Group, Malware and Malware Family, harmful software and its kind, and vulnerabilities and their kind. "Cause" refers to the relationship between vulnerabilities and the fallout from an attack. The relationship between operating systems, software, and hardware. As far as we are aware, no model has attempted to assess knowledge graph quality while being limited by low-dimensional embeddings. While the current models for knowledge representation have performed exceptionally well with high-dimensional embeddings in many downstream tasks, they still perform poorly with low-dimensional embeddings, which poses scalability issues in large-scale datasets.

The burgeoning interest in information extraction technology has led to the emergence of two primary knowledge extraction methods. The first method, rooted in knowledge engineering, heavily relies on extraction rules. While adept at handling information extraction within specific domains, this approach necessitates the involvement of domain experts and linguists during system development. Consequently, early information extraction systems predominantly adopt this rule-based approach due to its high accuracy rate. For instance, Pinkston et al. [22] devised a system leveraging ontology to extract attack information, drawing from extensive intrusion data for classification. Rehman and Mustafa [23] similarly employed a rule-based system to extract vulnerability information from text descriptions, utilizing TFIDF scores. Additionally, Lowis and Accorsi [24] employed string matching on CVEs to categorize service-oriented architecture (SOA) vulnerabilities, emphasizing accuracy over machine learning techniques. Despite its precision, the rule-based method necessitates significant human involvement in system development.

However, an alternative extraction strategy becomes necessary when traditional rule-based approaches are unable to identify patterns. Machine learning is the foundation of the second main strategy. An information-extraction model is first thoroughly trained on a sizable dataset. This model can subsequently extract relevant information without the need for predefined rules, though it demands an ample amount of training data for optimal performance. These repositories include detailed information about vulnerabilities, including names, descriptions, priorities, methods of exploit, related attributes, and more. Both Chinese and US vulnerability databases adhere to a standardized naming convention, facilitating seamless integration of vulnerabilities across different platforms. This standardization significantly enhances the exchange of vulnerability data. An archive of recorded attacks that include details on attack names, types, protocols, attributes, descriptions, degrees of severity, and

other pertinent information is called the Attack Rule Base. The Snort attack rule base is particularly noteworthy since it is a well-organized repository with each attack rule kept as a single line in a file for effective information organization and accessibility. It also makes it easier to identify logical conflicts inside knowledge graphs. On the other hand, statistical-based reasoning methods leverage machine learning techniques to derive new entity relationships from knowledge graphs based on statistical rules. It isn't able to do type reasoning across different datasets, though. Alternative techniques, like [25] and the Linked Hypernyms Dataset (LHD) [26], are restricted to their individual repositories and rely on certain patterns within structured text data to extract instance types.

ILP-based and ARM-based approaches are two more subcategories of schema induction techniques. ILP-based techniques use logic programming and machine learning to infer logical conclusions from case and background data. Lehmann et al. [27], for instance, presented a technique for learning description logic that gradually refines ideas from the most general to the most specific by using a downward-refined operator. This method was expanded upon in [28] to manage extensive semantic data sets such as DBpedia. Concurrently, Völker and Niepert [29] presented a statistical approach to generate conceptual linkages from knowledge maps by building transactional tables with SPARQL queries and extracting related conceptual links using ARM-based approaches. Using negative-association rule-extracting strategies, Fleischhacker and Völker [30] further investigated non-conceptual links and produced thorough experimental results [31,32]. The complete cybersecurity knowledge base model proposed in this work is organized around five key components: idea, instance, relation, properties, and rules. Figure 2 shows the design of this cybersecurity knowledge repository. As Figure 2 illustrates, the architecture is composed of three separate ontologies: attack, vulnerability, and assets.

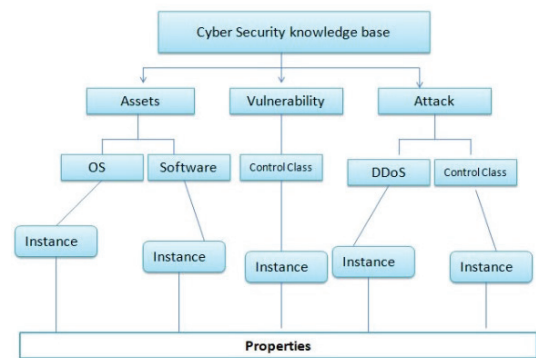


Fig. 2. The architecture of the cybersecurity knowledge base. DDoS, or distributed denial of service

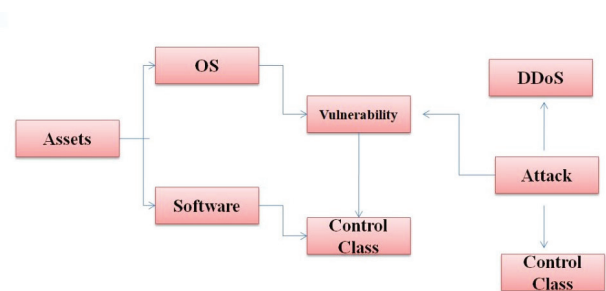


Fig. 3. The ontology of cybersecurity

V. RESULT AND DISCUSSION

For training NER1, initially, we chose attributes without gazettes. As Table 1 illustrates, NER1's average recognition performance for the 'consequence' object shows a noticeably high precision. In addition, the F1 measure shows that the recognition rates of "vulnerability" and "software" are positively correlated and greater than those of other entity sorts. This implies that the 'software' and 'vulnerability' recognition performance is praiseworthy, adding to an overall successful recognition final result. Then, we added features like gazettes, using clean Gazette as our choice for NER2 training. Furthermore, features that featured gazettes with sloppy Gazette as an option were used to train NER3. Table 2 provides specifics about the average recognition results.

TABLE I. RECOGNITION OUTCOMES FOR NER1

Entity	Precision	Recall	F_1
Vulnerability	0.835	0.701	0.721
OS	0.781	0.702	0.719
Total	0.762	0.751	0.716
Attack	0.851	0.601	0.703
Software	0.697	0.802	0.757

TABLE II. NER2 AND NER3 RECOGNITION RESULTS.

Model	Entity	Precision	Recall	F_1
NER2 (clean Gazette)	OS	0.602	0.821	0.754
	Attack	0.838	0.512	0.632
	Vulnerability	0.702	0.578	0.635
	Software	0.753	0.789	0.783
	Total	0.752	0.741	0.748
NER3 (sloppy Gazette)	Attack	0.821	0.483	0.612
	OS	0.786	0.851	0.811
	Vulnerability	0.726	0.587	0.639
	Software	0.821	0.789	0.801
	Total	0.801	0.751	0.785

Table 2 presents NER2's recognition accomplishments and shows that the organization does exceptionally well overall in identifying "software." Conversely, NER3 exhibits high F1 value in recognizing 'OS'. Notably, in terms of 'software' and 'OS' recognition, NER3 outperforms NER2, suggesting that the sloppy Gazette option contributes to improved recognition of cybersecurity-related entities. However, both NER2 and NER3 still exhibit low F1 measures for 'consequence' and 'mean', both falling below 70%. A comparative analysis of the three models is visually presented in Figure 8. Figure 8 shows that for the recognition of "software" and "OS," NER2 and NER3 perform better than NER1 in terms of accuracy, recall, and F1 measure. This emphasizes how important it is to use gazettes in NER training in the field of cybersecurity. However, achieving high accuracy for 'consequence' and 'mean' entities remains challenging without the use of gazettes for entity dictionary construction.

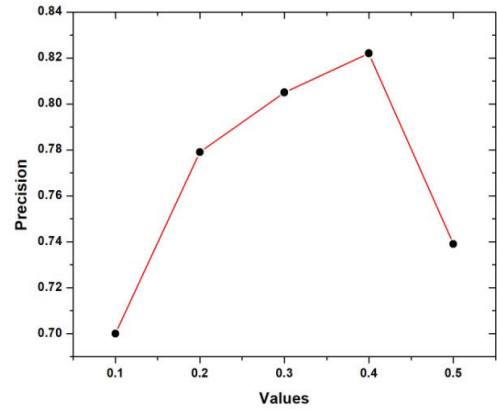


Fig. 4. Recognition results of NER1 Precision

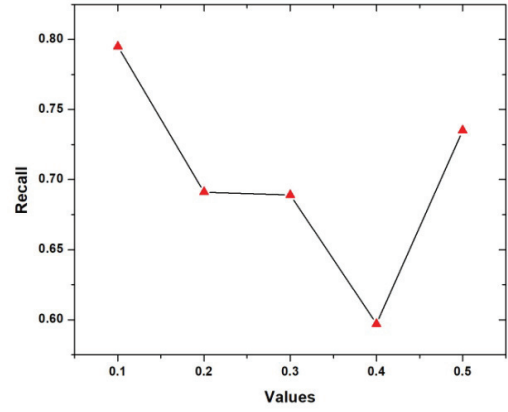


Fig. 5. Results of NER1 Recall recognition

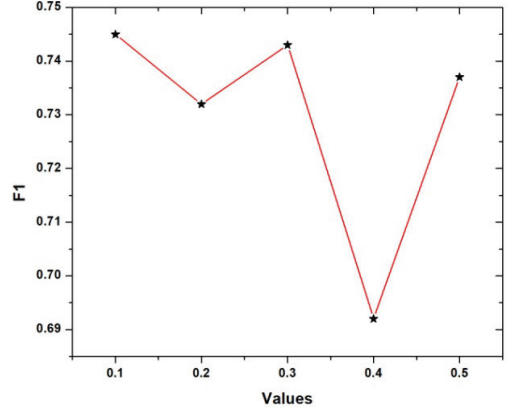


Fig. 6. Recognition results of NER1 F1

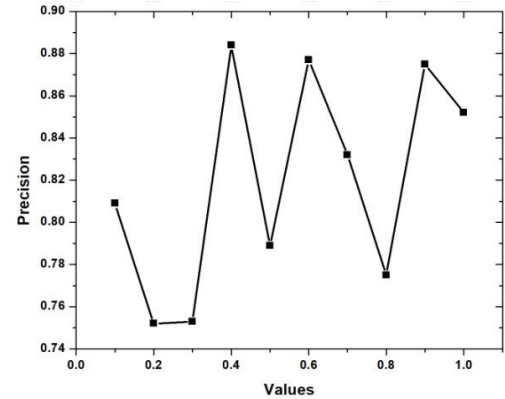


Fig. 7. Precision of NER2 and NER3 Recognition Results

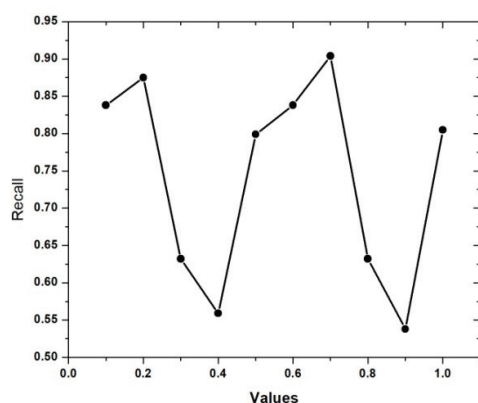


Fig. 8. Fig. 8. Recall outcomes for NER2 and NER3 recognition

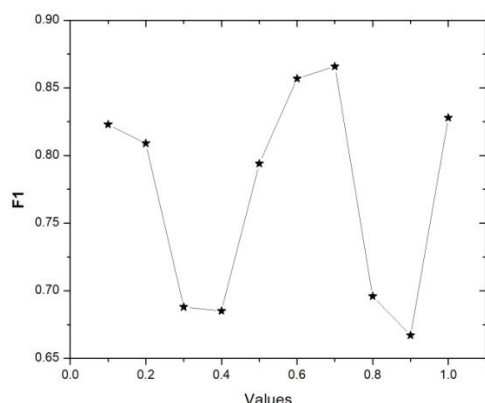


Fig. 9. Results of NER2 and NER3 F1 recognition

VI. CONCLUSION

This paper represents a significant contribution to the ongoing quest for effective cybersecurity solutions, presenting a comprehensive framework for building ontology-based knowledge systems tailored to the intricacies of cybersecurity domain. At its core, the framework revolves around the concept of vulnerability, recognizing it as a central pivot around which cyber threats revolve and defensive strategies coalesce. By anchoring the ontology in vulnerability, researchers can capture the multifaceted nature of cyber risk, encompassing both technical vulnerabilities in software and hardware systems as well as human factors such as social engineering and insider threats.

The development of the cybersecurity ontology represents a foundational step towards enhancing our understanding of cyber threats and vulnerabilities. By organizing knowledge into a structured framework of concepts, entities, and relationships, the ontology provides a common language for stakeholders to communicate and collaborate effectively. Moreover, it serves as a repository of domain-specific knowledge, capturing insights gleaned from empirical research, expert opinions, and real-world cyber incidents. In doing so, the ontology lays the groundwork for more sophisticated analysis and decision-making, empowering defenders to anticipate, mitigate, and respond to cyber threats with greater agility and efficacy.

In conclusion, the development of an ontology-based knowledge system for cybersecurity represents a critical step towards bolstering digital defenses and safeguarding critical assets in an increasingly interconnected world.

VII. FUTURE SCOPE

Threat Intelligence and Information Sharing: Ontology-based knowledge systems can facilitate the aggregation, analysis, and dissemination of threat intelligence across organizational boundaries, enabling more effective collaboration and information sharing among cybersecurity stakeholders. Future research efforts should focus on developing standards and interoperability frameworks for exchanging threat intelligence using ontological representations, fostering a more unified and coordinated approach to cyber defense.

Cybersecurity Education and Training: Ontology-based knowledge systems can also serve as powerful tools for cybersecurity education and training, providing learners with structured access to domain knowledge and real-world cyber scenarios. Future work in this area should explore the development of educational ontologies tailored to cybersecurity domain, incorporating interactive learning resources, simulations, and practical exercises to enhance learners' understanding of cyber threats and defensive strategies.

Cyber Risk Management and Compliance: Ontology-based approaches can assist organizations in managing cyber risk and ensuring compliance with regulatory requirements and industry standards. By formalizing knowledge about vulnerabilities, threats, and controls within the ontology, organizations can develop more effective risk assessment methodologies and compliance frameworks. Future research efforts should focus on developing ontological representations of cyber risk management processes, integrating them with existing risk assessment tools and compliance management systems to enhance their effectiveness and scalability.

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